

USN

--	--	--	--	--	--	--	--	--	--

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Sixth Semester B.Tech (CBCS) Degree Examinations
May 2025

EC4212-1- IMAGE PROCESSING AND FEATURE ENGINEERING
(For EC)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. Process in which the known data is utilized to evaluate the value at an unknown location is
 - A) Acquisition
 - B) Interpolation
 - C) Pixelation
 - D) Filtration
2. Output of a smoothing, linear spatial filter is _____.
 - A) Median of pixels
 - B) Maximum of pixels
 - C) Minimum of pixels
 - D) Average of pixels
3. Given an intensity level $[0, L-1]$ with "r" and "s" positive values, how will the negative of an image obtain?
 - A) $s = L - 1 - r$
 - B) $s = L - 1 + r$
 - C) $s = L + 1 - r$
 - D) $s = L + 1 + r$
4. In the visible spectrum the _____ color has the maximum wavelength.
 - A) Violet
 - B) Blue
 - C) Red
 - D) Yellow
5. Median filters belong to _____ category of filter.
 - A) Frequency Domain Filter
 - B) Order Static Filter
 - C) Linear Spatial Filter
 - D) Sharpening Filter
6. The difference in intensity between the highest and the lowest intensity levels in an image is _____.
 - A) Noise
 - B) Contrast
 - C) Saturation
 - D) Brightness
7. Two regions are said to be _____ if their union forms a connected set.
 - A) Adjacent
 - B) Disjoint
 - C) Closed
 - D) None
8. What is the process of moving a filter mask over the image and computing the sum of products at each location called as?
 - A) Correlation
 - B) Non-linear filtering
 - C) Linear filtering
 - D) Convolution
9. Response of derivative mask is zero at
 - A) Constant intensities
 - B) Sharp intensities
 - C) Low intensities
 - D) High intensities
10. Thresholding gives _____.
 - A) Large image
 - B) Gray-scale image
 - C) Binary image
 - D) Color image

11. Log function is also called as _____.
 A) Gaussian B) Gray-scale image
 C) Gradient image D) Mexican hat
12. Quantity that is used to measure the total amount of energy flowing from the light source is
 A) Brightness B) Intensity
 C) Luminance D) Radiance
13. Which of the following is not a type of feature engineering?
 A) Feature extraction B) Feature transformation
 C) Feature elimination D) Feature selection
14. Which variable type represents a countable and finite set of values?
 A) Continuous B) Discrete
 C) Categorical D) Ordinal
15. Which of the following is not a segmentation technique?
 A) Thresholding B) Edge detection
 C) Fourier Transform D) Region-based segmentation
16. The watershed algorithm is useful for segmenting _____.
 A) color images B) low resolution images
 C) binary images D) overlapping objects
17. Which of the following is true in the case of morphological dilation?
 A) It increases the brightness of the image B) It smooths boundaries and connects broken components
 C) It always removes foreground pixels D) It can be applied on both color and gray scale images
18. The primary goal of image segmentation is _____.
 A) Compress an image B) Convert an image to grayscale
 C) Separate objects from the background D) Enhance image contrast
19. Which morphological operations are used in opening?
 A) Dilation followed by erosion B) Two consecutive dilations
 C) Erosion followed by dilation D) Two consecutive erosions
20. Log transformation is primarily used to _____.
 A) reduce the effect of outliers B) normalize a dataset
 C) improve model accuracy D) increase dataset complexity

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit - I

1. a) Explain the three types of distance measures with relevant mathematical equations.
 b) Apply histogram equalization on the image shown in Fig. 1b. Sketch the original and equalized histograms.

4	4	4	4	4
3	4	5	4	3
3	5	5	5	3
3	4	5	4	3
4	4	4	4	4

Figure. 1(b)

2. a) Explain the fundamental steps in digital image processing with necessary block diagram.
 b) Compute the median value of the marked pixels shown in Fig. 2b, using a 3X3 mask.

18	22	33	25	32	24
34	128	24	172	26	23
22	19	32	31	28	26

Figure. 2b

-2-

Marks	BT*	CO*	PO*
8	L*2	1	1
8	L3	2	1
8	L2	1	1
8	L3	2	1

3. a) Compute D_4 , D_8 , and D_m distances between p and q for the image segment shown in Fig. 3a. Given that (i) $V = \{1, 2\}$, (ii) $V = \{0, 1\}$

3	1	2	1
0	2	0	2
1	2	1	1
1	0	1	2
(p)			(q)

Figure. 3a.

- b) Explain the following color models with appropriate geometric representations.
i) RGB ii) CMYK iii) HSV

8	L3	1	1
8	L2	2	1

Unit – II

4. a) Find the dilated version of the input image shown in Figure. 4a(1) using structuring element shown in Figure. 4a(2).

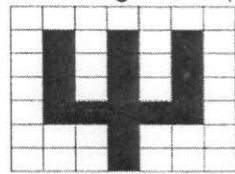


Figure. 4a(1)

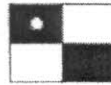


Figure. 4a(2)

- b) Explain an algorithm for optimum global thresholding method using Otsu's method.

8	L3	3	1
8	L2	4	1

5. a) Explain canny and Sobel edge detection methods with necessary filter masks.
b) Apply split and merge technique on the image shown in Figure 5(b) and draw the quad tree representation for the same.

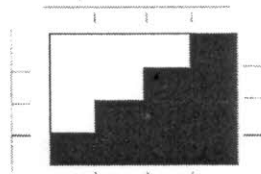


Figure 5(b)

8	L2	3	1
---	----	---	---

6. a) Explain the following morphological operations.
i) Opening ii) Closing

8	L3	4	1
---	----	---	---

- b) Explain the segmentation by morphological watersheds.

8	L2	3	1
8	L2	4	1

Unit – III

7. a) What is variable discretization? Explain equal-width binning and equal frequency binning methods.

8	L2	5	1
---	----	---	---

- b) Explain the following issues in Datasets.

- (i) Missing data.
(ii) Outliers.

8	L2	5	1
---	----	---	---

8. a) Explain wrapper method of feature selection with relevant block diagram.
b) Explain one-hot encoding method with an example.

8	L2	5	1
8	L2	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
